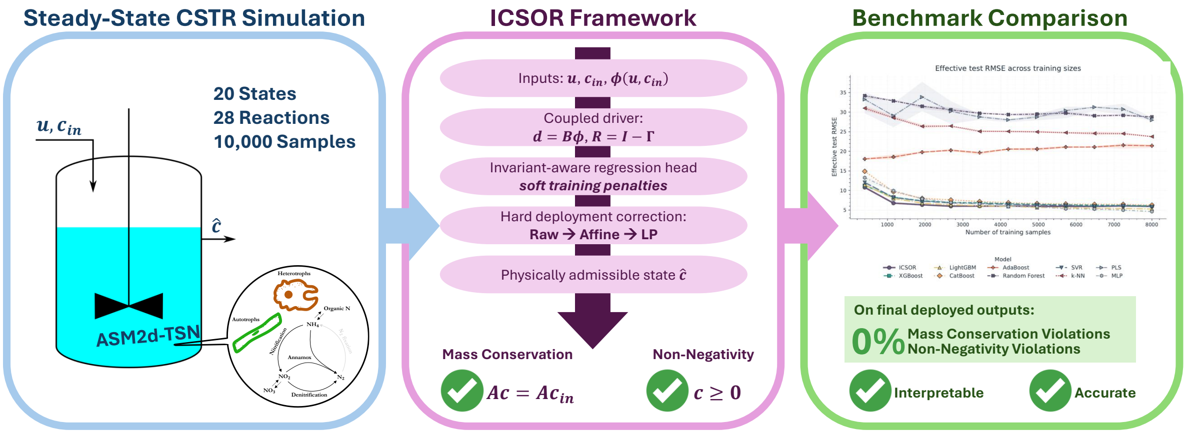
Graphical Abstract

An Interpretable Statistical Surrogate of Activated Sludge Systems that Preserves Mass Conservation and Component Non-Negativity



Highlights

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- ICSOR combines interpretable regression with constrained deployment.
- Deployed predictions satisfy conservation and non-negativity exactly.
- ICSOR gives the lowest mean RMSE at the smallest dataset size.
- ICSOR gives the lowest normalized RMSE area across dataset sizes.
- Physical guarantees involve a moderate fixed-split accuracy tradeoff.

An Interpretable Statistical Surrogate of Activated Sludge Systems that Preserves Mass Conservation and Component Non-Negativity

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ABSTRACT

Activated sludge models are essential for wastewater-treatment design and operation, but their detailed simulations can be too computationally demanding for repeated analysis. Machine-learning surrogates are faster, yet accurate predictions may still violate mass conservation or produce negative component concentrations. This study introduces Invariant-Constrained Second-Order Regression (ICSOR), an interpretable surrogate that combines a transparent, invariant-aware regression head with a constrained deployment procedure. The regression head predicts the effluent component state from influent conditions and operating inputs. At deployment, each prediction is checked, projected onto the mass-conservation constraints, and, when necessary, adjusted by a linear program to enforce both conservation and non-negativity. ICSOR was evaluated on 10,000 steady-state samples from an ASM2d-TSN continuous stirred tank reactor and compared with XGBoost, LightGBM, CatBoost, AdaBoost, Random Forest, support vector regression, k-nearest neighbors, partial least squares, and a multilayer perceptron. The deployed ICSOR predictions had zero conservation and non-negativity violations: the affine projection resolved 21.8% of test cases, and the linear program resolved the remaining 78.2%. For context, none of the raw regression-head predictions satisfied conservation, although 21.7% were non-negative, showing that the hard guarantee came from constrained deployment rather than training alone. This physical reliability involved a moderate accuracy tradeoff: aggregate test RMSE was 5.98, compared with 4.38 for the multilayer perceptron and 5.30 for LightGBM. Across the dataset-size study, however, ICSOR achieved the lowest normalized area under the RMSE learning curve and the smallest train–validation RMSE gap among the comparatively accurate models. ICSOR therefore offers a useful balance of interpretability, data efficiency, and guaranteed physical admissibility for steady-state activated sludge surrogate modeling.

Nomenclature

A	Full-row-rank invariant matrix defining the null space of the stoichiometric matrix
b	Baseline forcing vector of the blockwise driver ($b \in \mathbb{R}^F$)
B	Second-order driver coefficient matrix
c_{aff}	Affine invariant projection of the raw coupled state; exactly conservative but not necessarily non-negative
c_{in}	Influent ASM component concentration vector ($c_{in} \in \mathbb{R}^F$)
c_{out}	True steady-state effluent ASM component concentration vector ($c_{out} \in \mathbb{R}^F$)
c_{raw}	Unconstrained regression-head component prediction with no hard feasibility guarantee
$\hat{c}$	Final checked or corrected ICSOR component state returned at deployment
$\widehat{C}$	Fitted non-negative component prediction matrix used in training
D	Total dimension of the second-order feature map
F	Number of mechanistic ASM components
$\mathcal{G}$	Admissible set for Γ (zero diagonal, box bounds, conditioning threshold)
Γ	Learned ASM component coupling matrix
I_{comp}	Composition matrix mapping ASM components to measured composites ($I_{comp} \in \mathbb{R}^{K \times F}$)
K	Number of measured composite variables
$\kappa(\cdot)$	Matrix condition number
λ_B	Ridge regularization weight on B
λ_Γ	Ridge regularization weight on Γ
λ_{inv}	Soft invariant-residual penalty weight in the training objective
λ_{sys}	Coupled-system consistency penalty weight in the training objective

ORCID(s):

M_{op}	Number of operational variables
N	Number of steady-state training samples
N_{rxn}	Number of reactions in the mechanistic model
R	Coupled-system matrix, $R = I_F - \Gamma$
u	Operational input vector ($u \in \mathbb{R}^{M_{op}}$)
w_c	Componentwise weight vector used in deployed correction
y_{ext}	Reported measured effluent composite vector ($y_{ext} \in \mathbb{R}^K$)
$\hat{y}_{ext}$	Reported composite prediction, $\hat{y}_{ext} = I_{comp}\hat{c}$
ν	Stoichiometric matrix ($\nu \in \mathbb{R}^{N_{rxn} \times F}$)
ξ	Net reaction progress vector
Φ	Feature matrix collecting ϕ row-wise over the training set ($\Phi \in \mathbb{R}^{N \times D}$)
ϕ	Partitioned second-order feature map
Θ_{cc}	Influent–influent driver block ($\Theta_{cc} \in \mathbb{R}^{F \times F^2}$); each output-specific $F \times F$ reshape is symmetrized
Θ_{uc}	Operating–influent interaction driver block ($\Theta_{uc} \in \mathbb{R}^{F \times (M_{op} F)}$)
Θ_{uu}	Operating–operating driver block ($\Theta_{uu} \in \mathbb{R}^{F \times M_{op}^2}$); each output-specific $M_{op} \times M_{op}$ reshape is symmetrized
W_{in}	First-order influent driver block ($W_{in} \in \mathbb{R}^{F \times F}$)
W_u	First-order operating driver block ($W_u \in \mathbb{R}^{F \times M_{op}}$)
$\otimes$	Kronecker product

1. Introduction

Water pollution presents a critical environmental and public health challenge worldwide, with particularly severe repercussions in developing nations (Lin et al., 2022; Madhav et al., 2020). Biological wastewater treatment systems, particularly those based on the activated sludge process, are the dominant approach applied to combat this issue (Duan et al., 2022). However, their modeling and analysis present formidable engineering and computational challenges (Xiao et al., 2025). The underlying biophysical mechanisms are governed by the Activated Sludge Models (ASM) (Henze et al., 2006), which rely on coupled, multiplicative non-linear kinetic expressions to describe microbial growth and substrate consumption. This inherent non-linearity limits the practical application of transparent analytical methods for process design, frequently forcing engineers to rely on iterative trial-and-error approaches or computationally expensive heuristic algorithms (Aboagye et al., 2021; Padron-Paez et al., 2020).

In response to the computational and mathematical complexity inherent in non-linear Activated Sludge Models, research has increasingly focused on developing surrogate models to approximate the behavior of biological wastewater treatment systems (Bhosekar & Ierapetritou, 2018; Durkin et al., 2024; Tsochatzidi et al., 2025). Traditional approaches have explored linearization techniques (Li et al., 2020; Yang et al., 2014) and multivariable polynomial fitting (Ahn et al., 2014; Kim et al., 2009; Lim et al., 2011). While linear methods like Partial Least Squares (PLS) offer interpretable mappings, they inherently struggle to capture the severe non-linearities of microbial kinetics without extensive feature engineering; indeed, comparative wastewater studies report substantial error reductions when linear PLS is replaced by kernel-based or nonlinear inner-model variants (Liu et al., 2021; Woo et al., 2009; Yang et al., 2021). To overcome these limitations, the machine learning (ML) paradigm has emerged as a powerful route for approximating highly non-linear wastewater-treatment behavior (Croll et al., 2023; Nazlabadi et al., 2021). Recent wastewater evidence further shows that, when predictive accuracy alone is the criterion, modern ML models can outperform mechanistic ASM baselines for effluent nitrogen prediction (Wang et al., 2025). A wide array of data-driven algorithms—ranging from distance-based methods like k-Nearest Neighbors (k-NN) and PLS (Khatri et al., 2020; Wold et al., 2001), to kernel methods like Support Vector Regression (SVR) (Fang et al., 2011), tree-based ensembles such as Random Forest (RF), AdaBoost, XGBoost, LightGBM, and CatBoost (Sadri Moghaddam & Mesghali, 2023; Wang et al., 2025; Yang et al., 2025), and Artificial Neural Networks (ANN) (Asadi et al., 2017; Bagherzadeh et al., 2021; Mao et al., 2024)—have demonstrated strong predictive capabilities for effluent quality.

Despite their predictive accuracy, these standard ML algorithms remain structurally blind to a central physical requirement of activated sludge systems: material cannot be created or destroyed arbitrarily across the modeled control volume. A surrogate may fit observed effluent behavior well while still implying an impossible redistribution of the underlying ASM components, thereby violating the mass balances that govern COD, nitrogen, or phosphorus transfer.

Related composition-preserving ML studies show that even accurate tree-based predictors can exhibit persistent unphysical conservation residuals unless an explicit corrective mechanism is imposed (Sturm & Silva, 2025). At the same time, physically admissible concentration states must remain componentwise non-negative. These two requirements are inseparable in practice. A surrogate that conserves mass but predicts negative concentrations is still unusable, and a surrogate that preserves non-negativity while violating mass conservation remains physically misleading.

Recent scientific-ML literature shows that satisfying both requirements simultaneously is nontrivial. A substantial class of methods enforces conservation alone through conservative flux formulations, stoichiometric architectures, projection operators, Bayesian reconciliation, or completion strategies, but does not guarantee non-negativity of the corrected state (Beucler et al., 2021; Geiss et al., 2022; Hansen et al., 2024; Harder et al., 2022; Sturm & Silva, 2025; Sturm & Wexler, 2020, 2022). In that broader literature, completion methods reconstruct withheld variables to satisfy linear equality constraints (Beucler et al., 2021), Bayesian reconciliation treats conservation as a probabilistic update (Hansen et al., 2024), and recent atmospheric-composition studies use stoichiometric null-space constructions and invariant-restoring nudges to keep learned states on the conservation manifold (Sturm & Silva, 2025; Sturm & Wexler, 2020, 2022). Closely related soft-constraint approaches, such as physics-informed neural networks, can encourage physically plausible behavior but generally provide neither exact conservation nor strict positivity guarantees; more generally, soft-penalty conservation treatments need not drive residual violations to zero at deployment (Hansen et al., 2024; Raissi et al., 2019). Positivity-only output transformations—whether activation-function clipping, ReLU output layers, or manual thresholding—can prevent negative states, but without an accompanying conservation mechanism they do not by themselves enforce global material balance (Kircher & Votsmeier, 2025; Ruckstuhl et al., 2021; Utkarsh et al., 2025). Conversely, conservative projection steps that ignore positivity can still generate infeasible negative states in practice, which is precisely why joint enforcement remains difficult (Kircher & Votsmeier, 2025).

A smaller body of work has reported simultaneous enforcement of conservation and non-negativity. Kircher and Votsmeier (2025) develop positivity-preserving conservative corrections for mechanistic kinetics surrogates in atmospheric chemistry and heterogeneous catalysis, while Sturm et al. (2023) show in aerosol transport that mass-preserving scaling can maintain both total conservation and non-negativity for advected aerosol superspecies. Min et al. (2024) and Utkarsh et al. (2025) likewise propose hard-constrained learning frameworks that can impose equality and inequality constraints together in more general machine-learning settings. These results demonstrate that dual physical guarantees are achievable, but they arise in application settings whose state definitions, constraints, and deployment objectives differ substantially from those of activated sludge process surrogates.

This gap is consequential for wastewater engineering. Activated sludge models couple multiple conserved material inventories with biologically and chemically constrained component states, so violations of either mass conservation or non-negativity can distort interpretation, confound process optimization, and undermine confidence in surrogate-assisted design (Hansen et al., 2024; Liu et al., 2021; Straus & Skogestad, 2018; Tang et al., 2026). A physically constrained data-driven model would therefore offer several practical benefits: it would preserve the bookkeeping needed for COD, nitrogen, and phosphorus analysis, prevent impossible negative component concentrations, reduce reliance on ad hoc clipping or reconciliation (Ba-Alawi et al., 2021; Kircher & Votsmeier, 2025), and yield predictions that remain more defensible as operating conditions and influent loads vary. Yet, to our knowledge, prior wastewater surrogate and review literature has not reported an activated sludge surrogate that provides hard guarantees of both stoichiometric mass conservation and componentwise non-negativity within a single interpretable steady-state framework (Kircher & Votsmeier, 2025; Newhart et al., 2019; Wang et al., 2025).

This study addresses that need through Invariant-Constrained Second-Order Regression (ICSOR), an interpretable surrogate for steady-state activated sludge systems. ICSOR is designed for a practical task: using influent conditions and operating settings to provide a fast estimate of the effluent state that engineers can inspect and use with confidence. Unlike an unconstrained predictor, ICSOR checks the predicted component state before returning it and ensures that the final result conserves the modeled material inventories and contains no negative concentrations.

The method separates two responsibilities. First, a transparent regression model captures how operating conditions and influent loads are associated with the effluent components. Second, a deployment step checks physical admissibility and adjusts the prediction when needed. This arrangement preserves the speed and interpretability expected from a surrogate while preventing the impossible material balances and negative concentrations identified above.

ICSOR is intended to be understandable as well as physically dependable. Its fitted structure keeps the roles of operating conditions, influent loads, nonlinear effects, and relationships among effluent components visible. This differs from linear summaries such as PLS loadings and from local explanations of black-box neural networks. The transparent regression model supports process interpretation, while the separate deployment step is responsible for ensuring that the returned prediction is physically admissible.

The objective of this article is to evaluate whether ICSOR can provide useful steady-state predictions without sacrificing interpretability or the two essential physical requirements identified above. The study uses a standalone Continuous Stirred Tank Reactor and compares ICSOR with established unconstrained machine-learning models. It examines predictive accuracy, performance across dataset sizes, computational cost, and physical admissibility. The scope is limited to steady-state prediction: ICSOR guarantees mass conservation and non-negativity for the returned state, but it does not claim complete kinetic or thermodynamic feasibility and is not a replacement for dynamic simulation.

2. Theory and Calculation

2.1. Physical Scope, State Spaces, and Notation

A fixed reactor block or process unit represented by steady-state samples is considered. The system boundary used to define influent and effluent states is identical to the boundary used to define stoichiometric conservation. Any external source or sink that crosses that boundary must therefore be represented in the adopted stoichiometric model; otherwise, it lies outside the present conservation claim.

ICSOR is formulated natively in ASM component space. Its scientific output is the effluent component vector itself, not a measured aggregate. Measured composites are obtained only afterward through an external composition matrix. This choice is deliberate because mass conservation and componentwise non-negativity are both defined in ASM component coordinates.

Let M_{op} be the number of operational variables, F the number of ASM components, K the number of reported composite variables, and N_{rxn} the number of reactions. Single-sample vectors are written as column vectors. The following state spaces and operators are defined:

- $u \in \mathbb{R}^{M_{op}}$: Operational input vector (e.g., hydraulic retention time, aeration intensity).
- $c_{in} \in \mathbb{R}^{F}$: Influent ASM component concentration vector.
- $c_{out} \in \mathbb{R}^{F}$: True steady-state effluent ASM component concentration vector.
- $\hat{c} \in \mathbb{R}^{F}$: Final deployed ICSOR component prediction.
- $y_{ext} \in \mathbb{R}^{K}$: Reported measured composite vector.
- $I_{comp} \in \mathbb{R}^{K \times F}$: Composition matrix mapping ASM component concentrations to reported composites, such that $y_{ext} = I_{comp} c_{out}$.
- $\nu \in \mathbb{R}^{N_{rxn} \times F}$: Stoichiometric matrix.

The linear composition map I_{comp} remains external to the model itself. In the common wastewater-reporting case where its rows are entrywise non-negative, non-negative component predictions imply non-negative reported composites automatically. Thus ICSOR places the hard non-negativity guarantee where it is physically meaningful, namely on the predicted ASM component state.

2.2. Stoichiometric Structure and Conserved Quantities

Let $\nu \in \mathbb{R}^{N_{rxn} \times F}$ be the stoichiometric matrix written in the adopted ASM component basis. For a steady-state sample, define the net reaction progress vector $\xi \in \mathbb{R}^{N_{rxn}}$, scaled into concentration-equivalent units, such that the net change across the control volume is:

$$c_{out} - c_{in} = \nu^T \xi \tag{1}$$

The vector ξ is rarely observed directly. To eliminate it and isolate the conserved quantities, a full-row-rank matrix $A \in \mathbb{R}^{q \times F}$ is introduced whose rows span the left null space of ν, so that $A\nu^T = 0$. Multiplying the stoichiometric change relation by A yields:

$$A(c_{out} - c_{in}) = A\nu^T \xi = 0 \quad \Longrightarrow \quad Ac_{out} = Ac_{in} \tag{2}$$

This invariant-matrix construction follows the same mathematical idea used by Sturm and Wexler (2020) in mass-conserving photochemical surrogates, where a null-space matrix is used to preserve elemental inventories exactly. In the present setting, the conserved rows of A represent wastewater-material inventories implied by the adopted ASM reaction network and system boundary rather than atmospheric atom counts. Each row of A represents one independent conserved combination of ASM components implied by the adopted reaction network and system boundary. The basis of A is not unique, but once it is fixed, the invariant relation $Ac_{out} = Ac_{in}$ defines the exact conservation law that deployed ICSOR predictions must satisfy.

2.3. Coupled Second-Order Driver and Prediction-Space Training

Operational variables u alter the process environment, whereas influent concentrations c_{in} determine the material inventory presented to that environment (de Araujo et al., 2013; Machado et al., 2014). ICSOR keeps these roles explicit by using a partitioned second-order feature map:

$$\phi(u, c_{in}) = \begin{bmatrix} 1 \\ u \\ c_{in} \\ u \otimes u \\ c_{in} \otimes c_{in} \\ u \otimes c_{in} \end{bmatrix} \in \mathbb{R}^D \tag{3}$$

where $\otimes$ denotes the Kronecker product under column-wise vectorization and $D = 1 + M_{op} + F + M_{op}^2 + F^2 + M_{op}F$.

The feature map drives a latent component-space forcing vector rather than the final deployed prediction directly:

$$r(u, c_{in}) = B\phi(u, c_{in}) \tag{4}$$

Using a latent driver rather than predicting the deployed state directly is also consistent with stoichiometry-driven flux-to-tendency architectures such as the conservation-law neural-network formulation of Sturm and Wexler (2022). The present article adopts that separation in a different way: ICSOR keeps the driver explicitly linear-quadratic and interpretable, then combines it with a learned coupling matrix and a separate constrained deployment step rather than embedding a fixed stoichiometric layer inside a deep network.

Using the same partitioned basis, the driver can be written blockwise as

$$r(u, c_{in}) = b + W_u u + W_{in} c_{in} + \Theta_{uu}(u \otimes u) + \Theta_{cc}(c_{in} \otimes c_{in}) + \Theta_{uc}(u \otimes c_{in}), \tag{5}$$

with $b \in \mathbb{R}^F$, $W_u \in \mathbb{R}^{F\times M_{op}}$, $W_{in} \in \mathbb{R}^{F\times F}$, $\Theta_{uu} \in \mathbb{R}^{F\times M_{op}^2}$, $\Theta_{cc} \in \mathbb{R}^{F\times F^2}$, and $\Theta_{uc} \in \mathbb{R}^{F\times(M_{op}F)}$.

For output component k, let $Q_{uu}^{(k)} \in \mathbb{R}^{M_{op}\times M_{op}}$ and $Q_{cc}^{(k)} \in \mathbb{R}^{F\times F}$ denote the square reshapes of the corresponding rows of Θ_{uu} and Θ_{cc}. The full ordered Kronecker map retains both $x_i x_j$ and $x_j x_i$, so after every B update the implementation projects each self-quadratic slice onto the symmetric subspace,

$$Q \leftarrow \mathrm{sym}(Q) = \frac{Q + Q^T}{2}, \qquad Q \in \left\{Q_{uu}^{(k)}, Q_{cc}^{(k)}\right\}_{k=1}^{F}.$$

This projection preserves the coupled-QP optimum. For any vector x, $x^T Q x = x^T\,\mathrm{sym}(Q)x$, whereas

$$\|Q\|_F^2 = \|\,\mathrm{sym}(Q)\|_F^2 + \|\,\mathrm{skew}(Q)\|_F^2.$$

Consequently, removing the skew-symmetric part leaves every fitted value unchanged and cannot increase the ridge-regularized objective. Because the reported estimator uses $\lambda_B > 0$, the unique ridge solution has zero skew part, so the explicit projection makes the commutativity-induced equality $\theta_{k,ij} = \theta_{k,ji}$ hold to machine precision.

Cross-component dependence is introduced explicitly through a learned coupling matrix $\Gamma \in \mathbb{R}^{F\times F}$. In this study, Γ is estimated in an admissible set that fixes its diagonal to zero, bounds its off-diagonal entries, and rejects choices that make $R = I_F - \Gamma$ poorly conditioned. The coupled relation is written as

$$Rc = r(u, c_{in}), \qquad R = I_F - \Gamma \tag{6}$$

The sign of B is not constrained. That choice is deliberate: increasing and decreasing effects are both allowed in the latent driver, while non-negativity is enforced later where it matters physically, on the predicted component state itself.

Over a dataset of N steady-state samples, let $\Phi \in \mathbb{R}^{N\times D}$ collect the feature rows, let $C_{in} \in \mathbb{R}_+^{N\times F}$ collect influent component states, and let $C_{out} \in \mathbb{R}_+^{N\times F}$ collect effluent component targets. ICSOR introduces a fitted non-negative component prediction matrix $\widehat{C} \in \mathbb{R}_+^{N\times F}$ and estimates $(B, \Gamma, \widehat{C})$ through the coupled objective

$$\begin{aligned}(\widehat{B}, \widehat{\Gamma}, \widehat{C}) = \arg \min_{B,\Gamma\in\mathcal{G},\widehat{C}\in\mathbb{R}_+^{N\times F}} & \left\| C_{out} - \widehat{C} \right\|_F^2 \\ & + \lambda_{inv} \left\| (\widehat{C} - C_{in}) A^T \right\|_F^2 \\ & + \lambda_{sys} \left\| \widehat{C}(I_F - \Gamma)^T - \Phi B^T \right\|_F^2 \\ & + \lambda_B \|B\|_F^2 + \lambda_\Gamma \|\Gamma\|_F^2 .\end{aligned} \tag{7}$$

Each term has a distinct role. The first term fits effluent ASM components directly. The second softly penalizes invariant mismatch in component space; it is not a hard equality constraint. The third encourages consistency between the fitted states and the coupled driver relation. The ridge penalties stabilize the driver and coupling estimates, and the positive B ridge penalty also selects the symmetric minimum-norm representative of each duplicated ordered quadratic pair.

2.4. Deployment-Time Inference with Exact Conservation and Non-Negativity

Training determines $\widehat{B}$ and $\widehat{\Gamma}$. Deployment then predicts a new component state through a staged inference rule: a direct coupled evaluation, a closed-form affine projection, and, only when non-negativity is still violated, a final linear program. For a new sample $(u_*, c_{in,*})$, define

$$\phi_* = \phi(u_*, c_{in,*}), \qquad d_* = \widehat{B}\phi_*, \qquad \widehat{R} = I_F - \widehat{\Gamma}. \tag{8}$$

Whenever $\widehat{R}$ is nonsingular, the unconstrained coupled state is

$$c_{raw} = \widehat{R}^{-1} d_* . \tag{9}$$

A cheap affine presolve then restores exact invariants without yet enforcing non-negativity. The nearest invariant-consistent state to the raw predictor is

$$c_{aff} = \arg \min_{c\in\mathbb{R}^F} \|c - c_{raw}\|_2^2 \quad \text{subject to} \quad Ac = Ac_{in,*}. \tag{10}$$

Because A is full row rank, AA^T is invertible and the Lagrange conditions yield a closed-form solution:

$$c_{aff} = c_{raw} - A^T (AA^T)^{-1} A\,(c_{raw} - c_{in,*}). \tag{11}$$

The affine presolve is therefore the orthogonal projection of c_{raw} onto the invariant-consistent affine set $\{c : Ac = Ac_{in,*}\}$ and requires only a single matrix–vector multiplication at deployment. This closed-form correction plays the role of a conservation-restoring nudge: like the species-weighted correction proposed by Sturm and Silva (2025), it returns the nearest invariant-consistent state to the raw predictor. Here, however, the affine projection is treated as a cheap intermediate stage rather than the final deployed output, because exact invariants alone do not preclude negative ASM component concentrations.

If c_{raw} already satisfies the invariants and remains non-negative, ICSOR can deploy it directly. If c_{raw} is infeasible but c_{aff} is non-negative, the affine presolve is already sufficient. Only when the affine state still contains negative components does ICSOR activate the full correction program:

$$\begin{aligned} \hat{c} = \arg & \min_{c\in\mathbb{R}^F,\,\rho\in\mathbb{R}^F} w_c^T \rho \\ \text{subject to} \quad & Ac = Ac_{in,*}, \\ & c \geq 0, \\ & -\rho \leq c - c_{aff} \leq \rho, \\ & \rho \geq 0 \end{aligned} \tag{12}$$

with fixed componentwise correction weights $w_c \in \mathbb{R}_+^F$. Equation (12) is the final correction branch of the checked deployment rule. It enforces exact invariant preservation and componentwise non-negativity simultaneously while keeping the final state as close as possible to the coupled affine reference in a weighted L_1 sense. The hard feasibility guarantee is produced by this checked deployment rule, not by the training objective. In particular, the λ_{inv} term in Eq. (7) is a soft regularizer and does not guarantee that c_{raw} satisfies $Ac_{raw} = Ac_{in,*}$. A raw or affine state is returned only after it passes the applicable feasibility checks; otherwise Eq. (12) imposes $Ac = Ac_{in,*}$ and $c \geq 0$ jointly. Because the physically admissible $c_{in,*}$ is itself feasible under the adopted component basis, the LP feasible set is nonempty. That second stage reflects the same central lesson emphasized by Kircher and Votsmeier (2025): a conservative correction is only physically useful if positivity is preserved at the same time. Their scaled-space formulation is specifically designed to keep small predicted concentrations from crossing into negative values during correction. The present adoption targets steady-state activated sludge component predictions rather than atmospheric chemistry or heterogeneous catalysis, and it uses a weighted L_1 correction about the affine invariant reference rather than the exact scaling strategy proposed there.

Under the adopted non-negative component basis, $c_{in,*}$ is feasible whenever the influent state itself is physically admissible. The deployed problem is therefore not an afterthought but the mechanism by which ICSOR closes the following gap: every deployed prediction preserves the adopted conserved quantities and remains non-negative componentwise. The raw-check, affine-presolve, then LP-correction sequence is also practically important because it reserves the general optimizer for the subset of samples whose non-negativity constraints are genuinely active.

2.5. External Reporting and Blockwise Interpretability

If reported measured composites are needed after deployment, they are computed externally by

$$\hat{y}_{ext} = I_{comp}\hat{c}. \tag{13}$$

This collapse is not part of the regression model itself; it is a reporting operator applied after deployment. Keeping it external matters because different internal ASM component states can produce the same measured aggregates. The learned scientific object is therefore $\hat{c}$, not the reported collapse alone.

Interpretability is retained at two levels. Because ICSOR is constructed as an intrinsically transparent surrogate, process interpretation is taken from its explicit coefficient blocks and coupling terms rather than from a separate post hoc explainer (Rudin, 2019; Shen et al., 2023). First, the six blocks of B have direct engineering meaning in the latent driver: b is the baseline forcing, W_u and W_{in} encode first-order operating and influent effects, Θ_{uu} and Θ_{cc} encode within-subspace curvature, and Θ_{uc} encodes operation-loading interaction effects. Second, Γ describes how that driver is redistributed across ASM components through the coupled system matrix $R = I_F - \Gamma$. The deployed prediction must therefore be read as the result of a driver, a coupling structure, and a constrained inference solve acting together.

These coefficients are motivated by plant operation rather than by algebraic convenience. The operating block W_u corresponds to levers that engineers actually move, such as aeration intensity and hydraulic retention time, while W_{in} captures first-order loading carry-through from the influent state. The quadratic blocks then retain the regime dependence and interaction structure that plant interpretation requires: Θ_{uu} allows operating actions such as aeration to have different marginal effects across regimes, Θ_{uc} captures the fact that the same intervention acts differently under different influent loads, and Θ_{cc} captures coupled loading states such as simultaneous COD–nitrogen or phosphorus-related effects. In that sense, the block structure of B remains tied to practical diagnoses of nitrification loss, carbon limitation, and phosphorus breakthrough rather than to a purely algebraic expansion. More generally, recent hard-constrained learning frameworks likewise emphasize that nonlinear interaction structure must be retained if one wants expressive surrogates after equality and inequality constraints are imposed (Utkarsh et al., 2025).

The coupling coefficients in Γ serve a different purpose. They do not describe how the operating point generates forcing; they describe how that forcing is redistributed across ASM component balances once created. Couplings among dissolved oxygen and nitrifier channels, among ammonium, nitrite, and nitrate, among S_{PO4}, X_{PAO}, X_{PP}, and X_{PHA}, or between soluble and particulate COD-bearing states correspond to recurring material linkages long documented in activated-sludge nutrient-removal models for nitrification, denitrification, phosphorus storage, and hydrolysis-uptake behavior (Wentzel et al., 1992; Henze et al., 1999). The sign of a coupling coefficient indicates whether that linkage is reinforcing or compensating, and its magnitude indicates how strongly the redistribution must be represented to reproduce the observed state pattern. Persistent entries of Γ are therefore interpretable as recurring process linkages rather than as purely statistical artifacts.

2.6. Deterministic Recursive Coupled-QP Estimation and Modeling Boundaries

This article adopts the recursive coupled-QP training route rather than a generic stochastic first-order optimizer such as Adam. That choice is deliberate. Writing the training criterion in Eq. (7) as $J(B, \Gamma, \widehat{C})$, each restart performs deterministic block-coordinate Gauss–Seidel sweeps of the form

$$B^{(t+1,s)} = \arg\min_{B} J\left(B, \Gamma^{(t,s)}, \widehat{C}^{(t,s)}\right), \tag{14}$$

$$\Gamma^{(t+1,s)} = \arg\min_{\Gamma\in\mathcal{G}} J\left(B^{(t+1,s)}, \Gamma, \widehat{C}^{(t,s)}\right), \tag{15}$$

$$\widehat{C}^{(t+1,s)} = \arg\min_{\widehat{C}\in\mathbb{R}_{+}^{N\times F}} J\left(B^{(t+1,s)}, \Gamma^{(t+1,s)}, \widehat{C}\right). \tag{16}$$

After each full cycle, the objective is evaluated as

$$J^{(t+1,s)} = J\left(B^{(t+1,s)}, \Gamma^{(t+1,s)}, \widehat{C}^{(t+1,s)}\right). \tag{17}$$

For fixed $(\Gamma, \widehat{C})$, the B subproblem is a ridge-regularized linear least-squares solve followed by the prediction-preserving symmetric projection stated in Section 2.3. For fixed $(B, \widehat{C})$, the Γ subproblem is a convex box-constrained quadratic program. For fixed (B, Γ), the $\widehat{C}$ subproblem is a convex non-negative quadratic program in which invariant mismatch remains a soft quadratic penalty.

One practical reason is the matrix $R = I_F - \Gamma$ itself. Deployment requires R to remain invertible, and in practice it must remain well conditioned because both the raw coupled state and the subsequent correction depend on solving through R. A nearly singular R would amplify numerical noise, distort the affine presolve, and make the deployed correction unstable. The recursive coupled-QP route can control this requirement explicitly: each Γ update is box-constrained, its diagonal is fixed at zero, and the candidate is accepted only if $\kappa(I_F - \Gamma)$ stays below a prescribed conditioning threshold. If that threshold is violated, the candidate can be shrunk toward the origin and, if necessary, replaced by a safe fallback. Stochastic methods such as Adam, and indeed other generic first-order solvers, do not provide that admissibility guarantee by themselves. They may lower the objective while still visiting or returning nearly singular coupling matrices unless an additional conditioning projection is imposed externally.

This architecture is preferred here over generic stochastic first-order training for three reasons. First, under exact block minimization, each block update cannot increase the coupled objective, so the objective sequence within a restart is monotone non-increasing and converges in value to a stationary point under standard block-successive minimization conditions (Razaviyayn et al., 2013). Second, the optimization variables remain attached to interpretable convex subproblems instead of being absorbed into one optimizer trajectory with learning-rate, momentum, clipping, or epoch-count sensitivity. Third, with fixed initialization, restart policy, and solver tolerances, the recursive coupled-QP route is reproducible in a way that stochastic training need not be.

The solver choices follow the same principle of structural fidelity. OSQP is used for the convex QP subproblems because the Γ and $\widehat{C}$ blocks are sparse, box- and non-negativity-constrained quadratic programs (Stellato et al., 2018). The deployment problem in Eq. (12) is solved with HiGHS because, once c_{aff} is fixed, the deployed correction is a linear program rather than a quadratic one (Huangfu & Hall, 2018). A second computational advantage comes from the hierarchical deployment rule described in Section 2.4: the raw coupled state is checked first, the closed-form affine projection is evaluated second, and the LP is reserved only for states whose non-negativity constraints remain active after those cheaper steps. The manuscript therefore presents one definite ICSOR contract: direct component-space prediction, explicit learned coupling, invariant-aware training, exact constrained deployment, and deterministic recursive coupled-QP estimation.

The iterative estimator and the accelerated deployment logic can be summarized as follows.

Pseudocode 1. Deterministic recursive coupled-QP training

Input: Φ, C_{in}, C_{out}, A, solver settings
for restart $s = 1, \ldots, S$ **do**
  initialize admissible $\Gamma^{(0,s)}$ and non-negative $\widehat{C}^{(0,s)}$
  compute $B^{(0,s)}$ from the corresponding ridge solve and project every $Q_{uu}^{(k)}$ and $Q_{cc}^{(k)}$ onto the symmetric subspace
  for $t = 0, \ldots, T_{\max} - 1$ **do**
    update $B^{(t+1,s)}$ via Eq. (14) and project every $Q_{uu}^{(k)}$ and $Q_{cc}^{(k)}$ onto the symmetric subspace
    update $\Gamma^{(t+1,s)}$ via Eq. (15)
    enforce zero diagonal, box bounds, and $\kappa(I_F - \Gamma^{(t+1,s)}) \leq \kappa_{\max}$
    update $\widehat{C}^{(t+1,s)}$ via Eq. (16)
    compute $J^{(t+1,s)}$ via Eq. (17) and record its running minimum
    if the trailing objective slope is below tolerance **then stop the restart**
  end for
end for
retain the feasible restart with the smallest recorded objective
Output: $\widehat{B}$, $\widehat{\Gamma}$, $\widehat{C}$

Pseudocode 2. Hierarchical deployment with affine presolve

Input: u_*, $c_{in,*}$, $\widehat{B}$, $\widehat{\Gamma}$, A
form ϕ_* and compute the raw coupled state $c_{raw} = (I_F - \widehat{\Gamma})^{-1}\widehat{B}\phi_*$
if c_{raw} is invariant-consistent and non-negative **then return** c_{raw}
compute the closed-form affine invariant projection c_{aff} via Eq. (11)
if c_{aff} is non-negative **then return** c_{aff}
solve the deployment LP in Eq. (12) with c_{aff} as the reference state
return the projected state $\hat{c}$

3. Methodology

3.1. ASM2d-TSN Process Model, State Basis, and Reported Composites

The benchmark was generated from a standalone steady-state continuous stirred tank reactor based on Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN). The ASM2d-TSN implementation follows Guerrero et al. (2011), which was adopted as the basis for the process model; all stoichiometric values and simulation rate coefficients used here were taken from that study. The adopted process model contains $N_{rxn} = 28$ coupled biological and chemical processes spanning aerobic, anoxic, and anaerobic hydrolysis; aerobic heterotrophic growth on fermentable substrate and acetate; heterotrophic denitrification through nitrate and nitrite pathways; fermentation; storage, growth, and lysis pathways for phosphate-accumulating organisms (PAO); growth and lysis of ammonia-oxidizing bacteria (AOB) and nitrite-oxidizing bacteria (NOB); and precipitation-redissolution reactions for phosphorus-bearing metal solids. The state basis contains $F = 20$ ASM component states, comprising the dissolved variables S_O, S_F, S_A, S_{NH4}, S_{NO2}, S_{NO3}, S_{N2}, S_{PO4}, S_I, and S_{ALK} together with the particulate variables X_I, X_S, X_H, X_{PAO}, X_{PP}, X_{PHA}, X_{AOB}, X_{NOB}, X_{MeP}, and X_{MeOH}. The stoichiometric matrix, composition matrix, and the kinetic and stoichiometric parameter tables used for the adopted ASM2d-TSN implementation are provided in the Supplementary Excel file.

For every simulated steady state, the reported outputs were computed externally from the component state through the same composition matrix I_{comp} used throughout the study. The reported output basis therefore contains $K = 4$ composites: chemical oxygen demand (COD), total nitrogen (TN), total phosphorus (TP), and total suspended solids (TSS). The composite calculations are not unique to ICSOR; they are part of the benchmark-data construction used for every model family.

Table 1
Ranges of the 22 sampled inputs used in the Latin hypercube design.

Variable	Range	Unit	Variable	Range	Unit
HRT	[6.0, 36.0]	h	S_I	[10.0, 90.0]	g COD/m^3
Aeration	[0.5, 2.5]	–	S_{ALK}	[80.0, 260.0]	mol/m^3
S_O	[0.0, 0.5]	g O_2/m^3	X_I	[20.0, 120.0]	g COD/m^3
S_F	[20.0, 180.0]	g COD/m^3	X_S	[60.0, 280.0]	g COD/m^3
S_A	[5.0, 80.0]	g COD/m^3	X_H	[15.0, 100.0]	g COD/m^3
S_{NH4}	[12.0, 55.0]	g N/m^3	X_{PAO}	[5.0, 60.0]	g COD/m^3
S_{NO2}	[0.0, 3.0]	g N/m^3	X_{PP}	[2.0, 20.0]	g P/m^3
S_{NO3}	[0.0, 8.0]	g N/m^3	X_{PHA}	[1.0, 30.0]	g COD/m^3
S_{N2}	[0.0, 2.0]	g N/m^3	X_{AOB}	[0.5, 8.0]	g COD/m^3
S_{PO4}	[2.0, 18.0]	g P/m^3	X_{NOB}	[0.5, 8.0]	g COD/m^3
			X_{MeP}	[0.0, 12.0]	g TSS/m^3
			X_{MeOH}	[0.0, 12.0]	g TSS/m^3

Table 1 introduces the full Latin hypercube design used to generate the benchmark. The table is organized to distinguish the two manipulated operating variables from the soluble and particulate influent-component ranges, because those groups play different roles in the study: the operating variables define the reactor environment, whereas the influent components define the loading state presented to that environment. Read row-wise, the table gives the admissible interval for each of the 22 sampled inputs and therefore specifies the envelope within which the steady-state simulations, surrogate fitting, and later benchmarking claims are valid.

Within Table 1, the first two rows define the controllable reactor settings and the remaining rows define the influent composition envelope. The soluble-state bounds cover oxygen, carbon, nitrogen, phosphorus, inert, and alkalinity loadings, while the particulate-state bounds cover inert solids, biodegradable substrate, biomass pools, storage polymers, and precipitated solids. That separation matters later when interpreting ICSOR coefficients, because the model treats operating conditions and influent components as distinct blocks in both feature construction and physical correction.

3.2. Simulation Protocol, Initialization, and Steady-State Acceptance

The 22-dimensional sampling domain, consisting of the two operating variables and the 20-component influent state, was explored by Latin hypercube sampling with random seed 42 (McKay et al., 1979). Sampling continued until 10,000 valid steady states had been accepted. The complete accepted simulation dataset is provided as a Supplementary CSV file. At every sampled operating-influent point, the steady-state ASM balance equations were solved under box constraints by nonlinear least squares with lower and upper state bounds of 10^{-8} and 1500, variable, residual, and gradient tolerances of 10^{-9}, and a maximum of 1200 function evaluations. Up to 25 candidate attempts were permitted per target sample so that infeasible operating-influent draws could be replaced without changing the prescribed sample count.

The nonlinear solve was supplied with two physically informed initializations. The primary initialization blended the current influent state with the previous accepted steady state, promoted biomass pools with comparatively slow decay and demonstrated resilience across nearby operating conditions (Wentzel et al., 1992; Yilmaz et al., 2007), and estimated dissolved oxygen from the hydraulic-dilution and oxygen-transfer balance. A second multistart initialization then applied stronger substrate depletion and stronger biomass enrichment to probe alternative feasible basins. Both initializations were clipped to the admissible state bounds before optimization. If neither initialization yielded a solution whose maximum absolute residual was no greater than 10^{-9}, the model was dynamically relaxed for 20 days with a stiff backward differentiation formula integrator (Curtiss & Hirschfelder, 1952) and the terminal state of that dynamic relaxation was reused as a new steady-state guess. Only converged steady states that satisfied the residual-acceptance threshold were retained, so the final dataset contains only biologically viable operating points.

Table 2 summarizes these two initialization routes in the exact order they were applied. The table should be read as a decision aid for the nonlinear solver: the primary column defines the first physically informed guess used at each sampled operating point, and the multistart column defines the more aggressive fallback used to probe an alternative feasible basin before dynamic relaxation is invoked. Each row isolates one part of that initialization logic so that the steady-state acceptance protocol can be reproduced without ambiguity.

The structure of Table 2 reflects the intended solver behavior. The base-state row carries forward information from the previous accepted steady state when that information exists, which improves continuity across neighboring Latin hypercube samples. The substrate and biomass rows then bias the initialization toward biologically plausible regimes by

Table 2
Initialization heuristics used for the steady-state nonlinear solve.

Quantity	Primary initialization	Multistart initialization
Base state	$0.55\, c_{out}^{(prev)} + 0.45\, c_{in}$ when a previous accepted state exists; otherwise influent-based	Primary initialization reused as base state
S_A and S_F	70% of the base value	35% of the primary value
X_H	$\max(\text{base}, 0.25\, X_S)$	$\max(\text{primary}, 0.45\, X_S)$
X_{PAO}	$\max(\text{base}, 1.2\, X_{PP})$	$\max(\text{primary}, 1.8\, X_{PP})$
X_{AOB}	$\max(\text{base}, 0.03\, S_{NH4})$	$\max(\text{primary}, 0.08\, S_{NH4})$
X_{NOB}	$\max(\text{base}, 0.10\, S_{NO2})$	$\max(\text{primary}, 0.25\, S_{NO2})$
S_O	$\dfrac{D\, S_O^{(prev)} + k_L a\, S_O^{sat}}{D + k_L a}$ with $D = 24/\text{HRT}$, $k_L a = 2.0 + 18.0\,\text{Aeration}$, and $S_O^{sat} = 8.5$	Inherited from the primary initialization
Bounds	Clipped to $[10^{-6}, 1500]$	Clipped to $[10^{-6}, 1500]$

partially depleting readily consumed substrates and preserving key biomass pools. The dissolved-oxygen row supplies a separate process-based estimate tied to hydraulic dilution and oxygen transfer, and the bounds row ensures that every initialization remains inside the admissible numerical domain before the nonlinear least-squares solve begins.

3.3. Constraint Operators, Benchmark Dataset, and Common Training Targets

The invariant operator A used by ICSOR was built from the same stoichiometric model that generated the dataset. Specifically, if $\nu \in \mathbb{R}^{28\times 20}$ denotes the ASM2d-TSN Petersen matrix, then A was obtained as a basis for the left null space of ν so that $A\nu^T = 0$ and hence $Ac_{out} = Ac_{in}$ for admissible steady states. The null-space basis was obtained by singular value decomposition of ν^T, after which the retained basis vectors were transposed into row form, rounded for numerical stability, thresholded to remove negligible coefficients, and normalized by the first non-zero coefficient in each row.

A single benchmark dataset was constructed from the accepted steady states. Every model used the same exact 22 physical inputs, namely hydraulic retention time, aeration level, and the 20 influent ASM component concentrations. Every model was trained against the same 20-component effluent targets in ASM component space, and the same external composition map was then used to compute COD, TN, TP, and TSS from those predicted component states. The benchmark therefore holds fixed the sampled operating conditions, the training targets, and the composite-reporting rule across all model families. The methodological distinction lies only in the internal structure used to estimate the component state: ICSOR predicts the 20-component effluent state under invariant and non-negativity constraints using c_{in} as the invariant reference in training and deployment, whereas the unconstrained regressors map the same 22-dimensional input directly to an unconstrained 20-component effluent prediction.

This common component-space training target determines the comparison protocol. The classical regressors were fitted without invariant-aware penalties, a coupled non-negative fitted-state variable, or a constrained deployment rule. One could append post-prediction treatments such as analytic projection, null-space completion, conservative reconciliation, Bayesian correction, or positivity-preserving adjustment (Beucler et al., 2021; Hansen et al., 2024; Kircher & Votsmeier, 2025; Min et al., 2024; Sturm & Silva, 2025; Sturm & Wexler, 2020, 2022; Utkarsh et al., 2025), but that protocol was not applied to the baselines here. ICSOR differs in two distinct respects that should not be conflated: its regression head is trained with soft invariant and coupled-system penalties and a non-negative auxiliary fitted state, while exact conservation and non-negativity are imposed only on the final state returned by its deployment correction.

3.4. Benchmark Models and Optuna Hyperparameter Design

The benchmark comprised ICSOR and nine unconstrained regressors: XGBoost, LightGBM, CatBoost, AdaBoost, Random Forest, Support Vector Regression, k-Nearest Neighbors, Partial Least Squares, and one fully connected multilayer perceptron (MLP). The MLP baseline contained three hidden layers of widths 128, 64, and 32, used logistic activation, and mapped the 22-dimensional input vector to the 20-component effluent state.

Figure 1 summarizes the retained MLP architecture so that its input stage, hidden layers, output stage, and dense inter-layer connectivity are visually distinguished from the other benchmark models. The Optuna configuration used to tune every retained model is summarized in Table 3. The selected ICSOR settings are then listed separately in Table 4 because that model carries the additional coupled and constraint-aware hyperparameters that do not appear in the unconstrained baselines. For the remaining baselines, Table 5 groups the selected settings for the tree-based regressors, Table 6 groups the kernel, instance-based, and linear regressors, and Table 7 reports the final MLP-specific

Table 3
Common Optuna design used for all retained models.

Setting	Value
Objective	Validation mean squared error
Optimization direction	Minimize
Sampler	Tree-structured Parzen estimator
Pruning policy	None
Study seed	42
Trials per model	100
Training-pool fraction for tuning	0.50
Validation fraction within tuning subset	0.20
Tuning-training rows	3,200
Tuning-validation rows	800
Timeout	None

configuration. Taken together, these tables document both the common tuning contract and the final model-specific settings used in the benchmark and repeated dataset-size study.

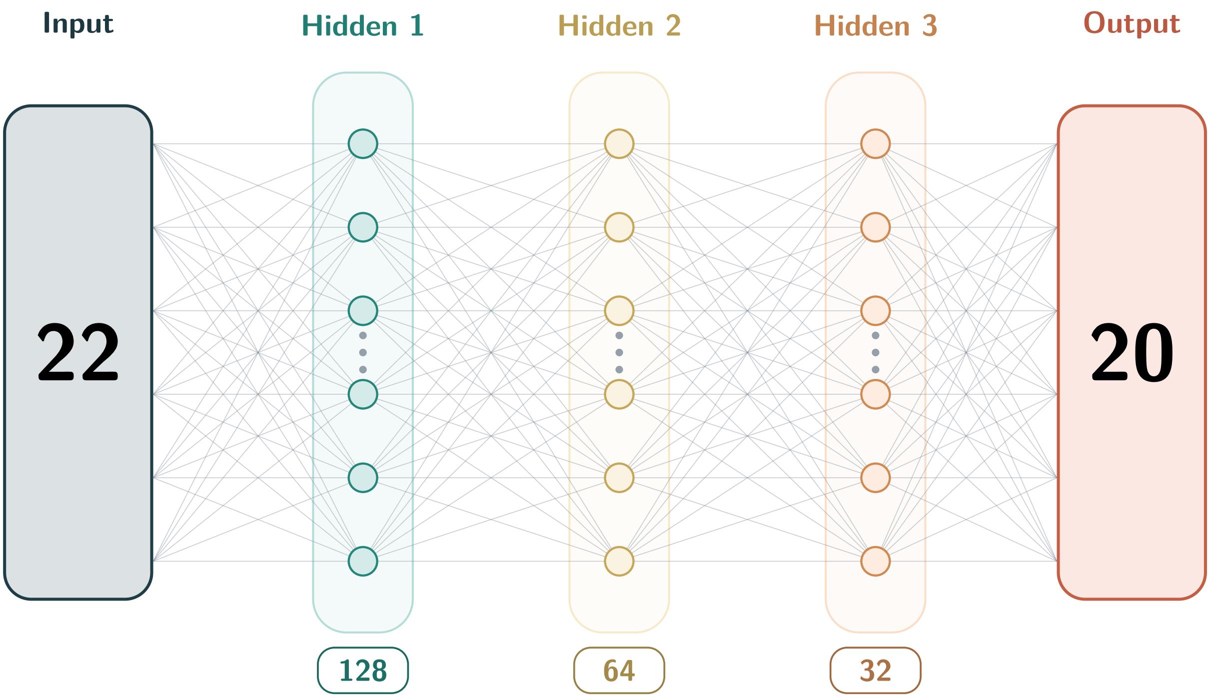

Figure 1: Architecture of the retained fully connected MLP baseline. The input and output stages are shown as labeled rectangular blocks to distinguish them from the hidden-layer neurons, which are shown as circular vertices. Representative vertices are displayed for the 22-input stage, the three hidden layers of widths 128, 64, and 32, and the 20-component effluent output stage, while the edge pattern indicates dense inter-layer connectivity between successive layers.

All retained models were assigned Optuna studies under a common tuning design (Akiba et al., 2019). Hyperparameter selection minimized validation mean squared error in the shared 20-component effluent prediction space using a seeded tree-structured Parzen estimator sampler (Bergstra et al., 2011), a no-pruning policy, and 100 trials per model. The authoritative 8,000-row training pool was first reduced to a 4,000-row tuning subset, corresponding to 50% of the training pool, and that subset was then partitioned with a 20% validation fraction, yielding 3,200 tuning-training rows and 800 tuning-validation rows. No wall-clock timeout was imposed. Once the Optuna studies were completed, the final selected hyperparameters were fixed and reused for the one-shot benchmark and the repeated dataset-size analysis.

Table 4
Optuna-selected ICSOR hyperparameters.

Parameter	Selected value
include_bias_term	true
λ_{inv}	8.68×10^{-3}
λ_{sys}	1.27×10^{-3}
λ_{B}	6.09×10^{-5}
λ_{Γ}	6.55×10^{-4}
γ_{abs}	0.368
max_outer_iterations	60
n_restarts	1
objective_regression_window	12
objective_regression_slope_tolerance	2.85×10^{-7}

Table 5
Optuna-selected hyperparameters for the tree-based regressors.

Model	Selected settings
XGBoost	$n_{estimators} = 309$, learning_rate= 0.125, max_depth= 3, min_child_weight= 6.84, subsample= 0.915, colsample_bytree= 0.828, reg_alpha= 1.18×10^{-3}, reg_lambda= 0.382
LightGBM	$n_{estimators} = 302$, learning_rate= 0.0809, num_leaves= 16, max_depth= 8, min_child_samples= 14, subsample= 0.845, colsample_bytree= 0.869, reg_alpha= 3.17×10^{-3}, reg_lambda= 3.36×10^{-4}
CatBoost	iterations= 236, learning_rate= 0.172, depth= 4, l2_leaf_reg= 0.0266, bagging_temperature= 0.0277, random_strength= 0.0657
AdaBoost	$n_{estimators} = 236$, learning_rate= 0.999, loss=square
Random Forest	$n_{estimators} = 390$, max_depth= 14, min_samples_split= 3, min_samples_leaf= 1, max_features= 0.500, bootstrap=false

Table 6
Optuna-selected hyperparameters for the kernel, instance-based, and linear regressors.

Model	Selected settings
SVR	$C = 0.118$, epsilon= 0.0458, kernel=poly, gamma=scale, degree= 4, tol= 8.96×10^{-4}, coef0= 0.889
k-NN	n_neighbors= 15, weights=distance, algorithm=kd_tree, leaf_size= 40, $p = 2$, metric=manhattan
PLS	n_components= 6, max_iter= 887, tol= 9.91×10^{-5}

Table 7
Optuna-selected hyperparameters for the retained fully connected ANN.

Parameter	Selected value
hidden_layer_sizes	(128, 64, 32)
activation	logistic
solver	adam
alpha	6.33×10^{-4}
batch_size	64
learning_rate_init	1.08×10^{-3}
max_iter	500
tol	1.63×10^{-5}
shuffle	false
early_stopping	false

3.5. Evaluation Protocol

A single row-wise 80–20 train-test split with random seed 42 was used for the one-shot benchmark, yielding 8,000 training rows and 2,000 test rows. The unconstrained regressors were standardized by z-score transformations fit on the training split only, separately for the 22-input feature matrix and the 20-output effluent-component matrix, to avoid preprocessing leakage from held-out data (Ba-Alawi et al., 2021; Pinheiro et al., 2025), and their predictions were inverse-transformed before external composite calculation and metric evaluation. ICSOR was kept in physical coordinates so that the invariant operator A, the coupled deployment rule, and the external reporting map I_{comp} remained aligned with the formulation of Section 2. For all models, COD, TN, TP, and TSS were computed from the predicted component states through the same composition map, after which aggregate R^2, root mean squared error, mean absolute error, and mean absolute percentage error and per-target root mean squared error were reported over those four composites.

For the predictive-accuracy comparison, ICSOR was evaluated at its final checked deployment output. The comparator regressors were evaluated after the benchmark's common equality-only affine alignment, followed by the same external composition map; this alignment enforces the reporting invariants but does not impose component non-negativity and is not the ICSOR LP. Raw-head physical-admissibility frequencies are reported separately so that this numerical alignment is not mistaken for a property learned by an unconstrained regressor.

Sample-efficiency, computational scaling, and ranking stability were examined through a repeated dataset-size analysis based on repeated random train-test evaluations at each retained size. The 10,000-sample benchmark was subsampled without replacement at total sizes of 500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, and 10,000 rows. At each retained total size, ten independent subsamples were drawn from the full benchmark using deterministic seeds derived from the base seed 42, and each subsample was split row-wise into an 80% training partition and a 20% held-out test partition using the same deterministic seed for that run. All ten retained models were trained and evaluated under the same size schedule, seed construction, and train-test fraction, so that observed performance differences at a given size reflect model structure rather than differences in protocol. This study contract yielded 110 train-test re-estimations per model across the full size sweep (11 retained sizes $\times$ 10 repeated runs per size). The Optuna-selected settings listed above were held fixed throughout this sweep so that the resulting learning curves isolated sample-size sensitivity and run-to-run variability rather than retuning noise. For every fit within this sweep, wall-clock training time was recorded, and computational scaling with dataset size was summarized from the distribution of training seconds across the ten repeated runs at each retained size. Predictive-performance ranking was based primarily on the mean held-out RMSE over the repeated runs, computed from the final reported composite outputs, and was supplemented by held-out MAE, MSE, MAPE, R^2, a normalized area-under-the-learning-curve summary, dense average rank across target-metric-size combinations, and the train-validation RMSE gap at the largest retained dataset size. Area-under-learning-curve summaries have prior use in the learning-curve and active-learning literature (Cawley, 2011; Guyon et al., 2011; Viering & Loog, 2023). Here, a simple linear normalization over retained dataset size was used. Writing $\bar{M}_{i,m}(n)$ for the mean held-out metric of model i at dataset size n and metric m over the repeated runs, the normalized area under the learning curve was computed by

$$\mathrm{nAUC}_{i,m} = \frac{1}{n_{\max} - n_{\min}} \int_{n_{\min}}^{n_{\max}} \bar{M}_{i,m}(n)\, dn, \tag{18}$$

evaluated numerically by the trapezoidal rule over the sampled dataset sizes. For each metric m, target k, and size n, models were assigned a dense rank $r_{i,m,k}(n)$ from the mean held-out metric over repeated runs, with lower-is-better ordering for RMSE, MAE, MAPE, and MSE and higher-is-better ordering for R^2; this rank-based aggregation follows the same general benchmarking logic used for cross-dataset average-rank summaries in machine-learning evaluation (Demšar, 2006; Shevchenko et al., 2024), and the metric-specific average rank and overall average rank were then

$$\bar{r}_{i,m} = \frac{1}{KS} \sum_{k=1}^{K} \sum_{s=1}^{S} r_{i,m,k}(n_s), \qquad \bar{r}_i = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \bar{r}_{i,m}, \tag{19}$$

where $\mathcal{M} = \{\mathrm{RMSE}, \mathrm{MAE}, R^2, \mathrm{MAPE}, \mathrm{MSE}\}$. At the largest retained dataset size $n_{\max}$, the train-validation RMSE gap was

$$\Delta_i^{\mathrm{RMSE}}(n_{\max}) = \overline{\mathrm{RMSE}}_i^{\mathrm{val}}(n_{\max}) - \overline{\mathrm{RMSE}}_i^{\mathrm{train}}(n_{\max}), \tag{20}$$

so positive values indicate that the training partitions outperformed the held-out test partitions.

Physical-admissibility diagnostics were evaluated before external composition collapse. To distinguish learned feasibility from post-processing, Table 11 uses the final checked ICSOR output but the direct raw-head output of each unconstrained comparator; Section 4.3 reports the raw, affine, and final ICSOR stages separately. For a component prediction $\tilde{c}$, mass-conservation violation was quantified by $v_{mc} = \|A(\tilde{c} - c_{in})\|_2$, and non-negativity violation was quantified by $v_{nn} = \|\min(\tilde{c}, 0)\|_1$, where the minimum is taken componentwise. A sample was counted as violating mass conservation or non-negativity when the corresponding quantity exceeded a numerical tolerance of 10^{-10}. For each model, violation frequency was reported as the percentage of test samples that violated each condition.

Table 8
Fixed-split 80–20 test performance in the reported-composite space. ICSOR is evaluated after its checked deployment rule; comparator predictions receive the benchmark's common equality-only affine alignment before external composition collapse.

Model	RMSE	MAE	R^2	MAPE
MLP	4.38	2.55	0.9915	0.0123
LightGBM	5.30	3.21	0.9806	0.0203
SVR	5.97	3.49	0.9713	0.0244
ICSOR	5.98	3.70	0.9700	0.0256
XGBoost	6.12	3.81	0.9733	0.0244
CatBoost	6.45	4.07	0.9727	0.0250
AdaBoost	21.27	13.15	0.8643	0.0653
k-NN	23.61	13.87	0.8535	0.0557
PLS	28.24	17.05	0.7808	0.0681
Random Forest	29.00	16.93	0.7934	0.0632

Table 9
Per-target fixed-split test RMSE for the retained models in the reported-composite space.

Model	COD	TN	TP	TSS
MLP	6.48	1.58	0.0361	5.66
LightGBM	7.59	2.77	0.0361	6.85
SVR	8.87	3.50	0.0361	7.20
ICSOR	7.77	3.55	0.0361	8.36
XGBoost	8.78	3.27	0.0361	7.88
CatBoost	9.13	3.23	0.0361	8.53
AdaBoost	38.30	5.58	0.0361	17.68
k-NN	38.24	4.23	0.0361	27.37
PLS	43.22	5.09	0.0361	36.00
Random Forest	45.87	4.00	0.0361	35.26

4. Results and Discussion

4.1. Overall Benchmark Performance on the Fixed Train-Test Split

Table 8 summarizes the fixed 80–20 benchmark after each model's predicted ASM component state has been collapsed through the same external composition map to COD, TN, TP, and TSS. Aggregate test RMSE is adopted as the primary ordering metric because it measures error in the same reported-composite space used throughout the comparison. On that criterion, the retained models separate into a leading group and a trailing group. The MLP attains the lowest aggregate RMSE (4.38), MAE (2.55), R^2 (0.9915), and MAPE (0.0123); LightGBM follows at RMSE 5.30; and SVR and ICSOR are closely spaced at 5.97 and 5.98, respectively.

In terms of predictive accuracy on the fixed split, the MLP attains the lowest error among the retained baselines and LightGBM attains the lowest among the tree-based methods. ICSOR does not attain the lowest aggregate RMSE: its final deployed outputs give 5.98, which is 36.5% higher than the MLP value of 4.38 and 12.8% higher than the LightGBM value of 5.30. ICSOR remains within 0.01 of SVR and below XGBoost and CatBoost. These results define a moderate accuracy–feasibility tradeoff. Under the stated reporting protocol, the comparator values include the common equality-only affine alignment but no LP non-negativity correction, whereas ICSOR is evaluated after its complete checked deployment rule.

Table 9 indicates that the aggregate ordering is driven primarily by COD, TN, and TSS. The MLP attains the lowest RMSE on all three of those composites, while LightGBM attains the lowest among the non-neural methods. On COD alone, ICSOR (RMSE 7.77) is 0.18 above LightGBM and below XGBoost, CatBoost, SVR, and the remaining baselines. Its larger TN and TSS errors, however, prevent that COD performance from translating into a correspondingly favorable aggregate rank. TP contributes negligibly to model differentiation: the equality-aligned reported outputs give an RMSE of 0.0361 for every retained method. The per-target decomposition therefore suggests that, once phosphorus prediction is effectively saturated, the observable model separation arises from carbon, nitrogen, and suspended-solids behavior.

Because the leading fixed-split models remain closely spaced on aggregate RMSE, a single train-test partition does not suffice to characterize sample efficiency or ranking stability. The repeated dataset-size study described below provides a broader basis for distinguishing models whose performance on one split may not generalize from those that remain competitive as the amount of training data varies.

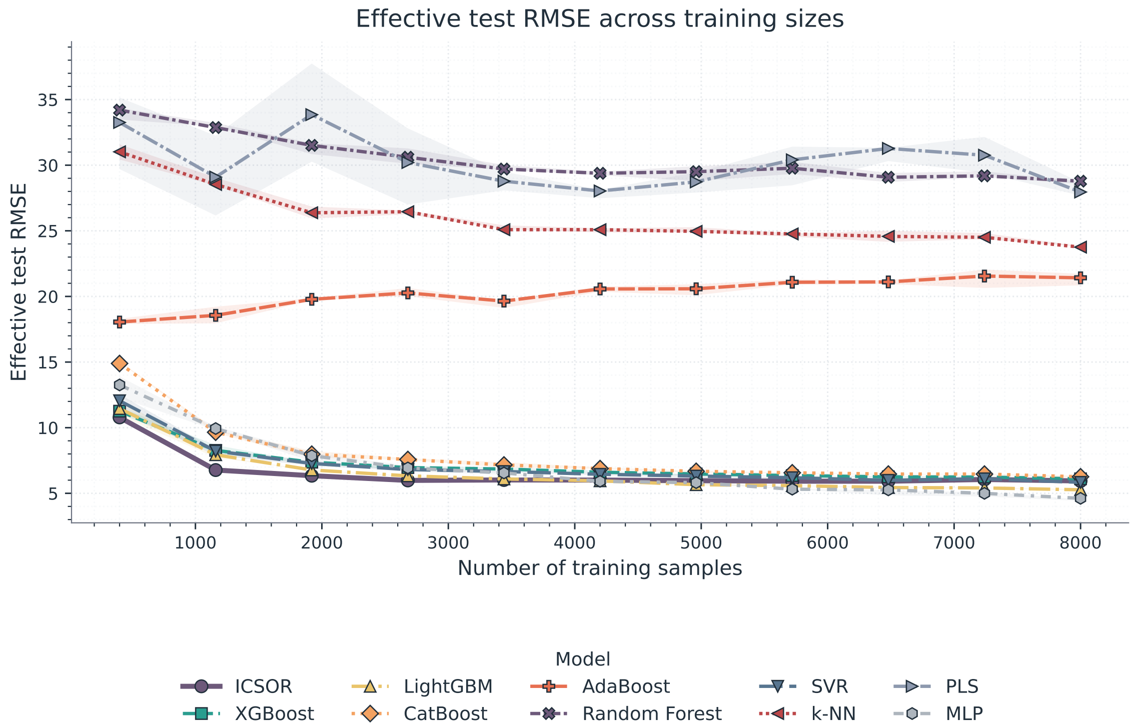

Figure 2: Repeated dataset-size learning curves from the repeated random train-test evaluations at each retained size. Lines show the mean held-out RMSE across the retained dataset-size sweep, and shaded bands show the interquartile range across the ten repeated runs at each size.

4.2. Sample Efficiency, Ranking Stability, and Generalization

Figure 2 shows a pattern that the fixed split alone does not capture: at the smallest retained dataset size of 500 rows, ICSOR attains the lowest mean held-out RMSE (10.79) among the retained models, followed by XGBoost (11.26) and LightGBM (11.45), while the MLP ranks fifth (13.26). As dataset size increases, however, the ordering shifts. By the full 10,000-row dataset size, the MLP attains the lowest RMSE (4.62), followed by LightGBM (5.27), SVR (5.84), and ICSOR (5.94). This crossover is consistent with broader learning-curve and hybrid-modeling literature, in which models with stronger structural priors can be more sample-efficient at small N, whereas more flexible unconstrained models may overtake them once sufficient data are available (Viering & Loog, 2023; Shen et al., 2023).

Table 10 condenses the repeated-size sweep into four summary statistics. ICSOR attains the lowest RMSE nAUC (6.33), indicating that its average RMSE profile across the full range of retained dataset sizes is favorable even though it does not attain the lowest final-size RMSE. LightGBM attains the lowest sample-average rank (2.2) and the MLP is second (2.5), indicating that when the evaluation is broadened across all reported metrics, targets, and retained dataset sizes, the unconstrained baselines with the lowest pointwise errors also tend to rank well on a broader set of criteria. ICSOR's sample-average rank of 4.3 is comparable to that of XGBoost (4.3), suggesting that its low-data advantage, while observable, does not translate into a dominant rank across the full evaluation grid.

The generalization behavior observed in Table 10 warrants separate attention. The train–validation RMSE gap of ICSOR at the largest retained dataset size is 0.22, lower than those of LightGBM (1.99), XGBoost (1.44), SVR (1.19), and the MLP (0.65). PLS also exhibits a small gap (0.38), but its training and held-out errors are both large, consistent with high-bias underfitting and prior evidence that linear PLS can mis-specify strongly nonlinear wastewater behavior (Geman et al., 1992; Woo et al., 2009). The repeated-size analysis therefore indicates comparatively small train–validation separation for ICSOR, while the fixed-split results in Section 4.1 still show the moderate absolute accuracy tradeoff relative to the MLP and LightGBM.

Figure 3 shows that this favorable generalization comes at a cost in training time. At the smallest retained dataset size of 500 rows, ICSOR requires a mean of 21.5 s, compared with 4.72 s for LightGBM, 3.16 s for XGBoost, and

Table 10
Summary of the repeated dataset-size benchmark. Final RMSE and ΔRMSE are evaluated at the largest retained dataset size, and lower values are better for all columns except that a smaller ΔRMSE indicates less train-validation separation.

Model	Final RMSE	RMSE nAUC	Sample Avg. Rank	ΔRMSE
MLP	4.62	6.75	2.5	0.65
LightGBM	5.27	6.35	2.2	1.99
SVR	5.84	6.90	4.0	1.19
ICSOR	5.94	6.33	4.3	0.22
XGBoost	6.10	7.01	4.3	1.44
CatBoost	6.26	7.60	5.0	0.69
AdaBoost	21.42	20.29	8.2	0.23
k-NN	23.76	25.77	7.0	23.66
PLS	27.96	30.17	9.4	0.38
Random Forest	28.78	30.31	8.2	18.71

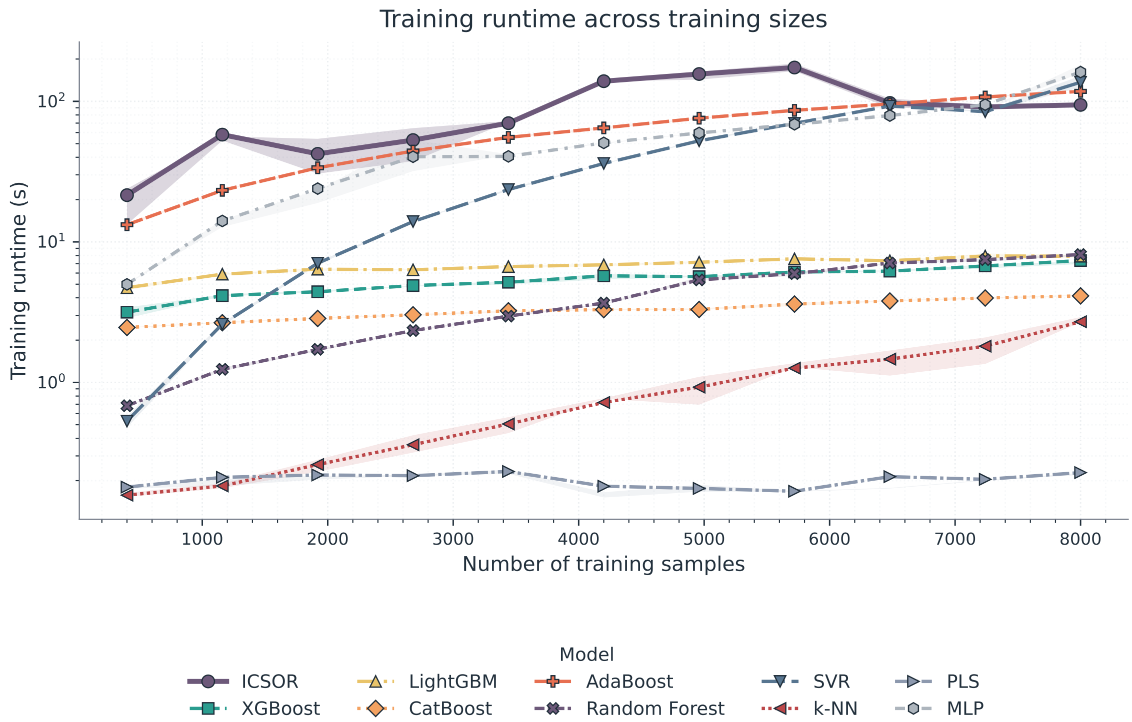

Figure 3: Repeated dataset-size runtime curves for the retained models. Lines show the mean training time across repeated runs and shaded bands show the interquartile range across the ten repeated runs at each size; the y-axis is logarithmic to keep the full cost spread readable.

4.99 s for the MLP. At the full 10,000-row dataset size, ICSOR rises to 94.4 s, which is slower than the tree-based models (4.13–8.13 s), PLS (0.228 s), and k-NN (2.70 s), but remains below AdaBoost (117.7 s), SVR (136.8 s), and the MLP (161.6 s). The recursive coupled-QP route is therefore more expensive than the fastest retained methods at every retained size, but it does not become the costliest option at the larger dataset scales and remains practical for offline surrogate fitting.

4.3. Physical Admissibility and Deployment Behavior

Table 11 reports the final checked ICSOR output alongside the direct raw-head outputs of the unconstrained comparators. The ICSOR deployment pipeline is the only retained pipeline that produces 0.0% mass-conservation violations and 0.0% non-negativity violations on the reported component-space test predictions. Every unconstrained baseline raw head violates conservation on 100% of the test samples. Non-negativity violation frequencies vary across models, ranging from 0.0% for AdaBoost, Random Forest, and k-NN to 70.1% for CatBoost, 68.65% for SVR, 64.95% for XGBoost, 58.2% for the MLP, 48.7% for PLS, and 46.0% for LightGBM.

Table 11
Physical-admissibility violation frequencies for the final checked ICSOR output and the direct raw-head outputs of the unconstrained comparators. Mass conservation and non-negativity are assessed before external composition collapse.

Model	Mass-Conservation Violations (%)	Non-Negativity Violations (%)
MLP	100.0	58.20
LightGBM	100.0	46.00
SVR	100.0	68.65
ICSOR	0.0	0.0
XGBoost	100.0	64.95
CatBoost	100.0	70.10
AdaBoost	100.0	0.0
k-NN	100.0	0.0
PLS	100.0	48.70
Random Forest	100.0	0.0

The zero non-negativity frequency for AdaBoost, Random Forest, and k-NN should not be interpreted as full physical admissibility. These regressors still violate the invariant conservation relations on every test sample, but their componentwise predictions are formed through aggregation rules that can preserve non-negativity when trained on nonnegative concentration targets: Random Forest averages nonnegative leaf responses, k-NN averages nonnegative neighbor responses, and AdaBoost combines nonnegative tree responses through a weighted-median ensemble rule. Their reported 0.0% non-negativity rate therefore reflects the geometry of these unconstrained predictors in the positive orthant, not enforcement of stoichiometric feasibility.

These diagnostics illustrate why predictive accuracy in the reported composite space does not by itself ensure physically interpretable component-level predictions. AdaBoost, Random Forest, and k-NN remain nonnegative on the test set, yet all three still violate the stoichiometric invariants on every sample; positivity alone therefore does not recover consistent material bookkeeping. Conversely, several of the models with the lowest aggregate RMSE, including the MLP and LightGBM, violate the conserved material inventories on every test sample and produce negative component concentrations on a substantial fraction of the test set. A surrogate may thus appear accurate in the reported composite space while producing component-level states that are not physically admissible, a mismatch that has also been documented in atmospheric-composition and reaction-kinetics surrogates when favorable aggregate error metrics obscure physically unacceptable species- or component-level behavior (Kircher & Votsmeier, 2025; Sturm & Silva, 2025).

The staged deployment logic can also be examined at the level of individual correction stages. None of the raw coupled test predictions was simultaneously invariant-consistent and non-negative: raw feasibility was 0.0%, raw conservation compliance was 0.0%, and 21.7% of raw states was non-negative. The closed-form affine projection restored invariants for all samples and was itself sufficient for 21.8% of the test set (436 of 2000 samples), resolving those cases without invoking a numerical optimizer. The remaining 78.2% (1564 samples) required the final LP correction, which converged in a mean of 23.5 HiGHS iterations with a maximum of 40. The final returned prediction was feasible for 100% of the test cases. Thus, the zero final violation rates are properties of the constrained deployment map, not of the learned regression head in isolation.

Taken together, these results show a moderate accuracy–feasibility tradeoff within the present benchmark, consistent with prior studies in which hard corrective constraints improve physical admissibility while sometimes sacrificing pointwise accuracy (Hansen et al., 2024; Sturm & Silva, 2025). When the sole objective is minimum reported-composite error, the MLP and LightGBM are preferable. When the objective also requires final predictions that satisfy invariant conservation and component non-negativity, the ICSOR constrained-deployment pipeline is the only retained pipeline that provides both guarantees.

4.4. Interpretable Raw-Head Structure of the Learned ICSOR Model

Coefficient interpretation is performed on the globally defined pre-correction head rather than on the piecewise deployment correction. Since $c_{raw} = \widehat{R}^{-1}\widehat{B}\phi$, its reported-composite counterpart is

$$y_{raw} = I_{comp}c_{raw} = B_{raw,ext}\phi, \qquad B_{raw,ext} = I_{comp}\widehat{R}^{-1}\widehat{B}. \tag{21}$$

Figure 4 presents the COD row's Θ_{cc} block from $B_{raw,ext}$. The COD, TN, TP, and TSS raw-head atlases are provided as Supplementary Figures S1–S4. Rows of $\widehat{B}$ are ASM component-driver rows and therefore cannot be indexed directly by

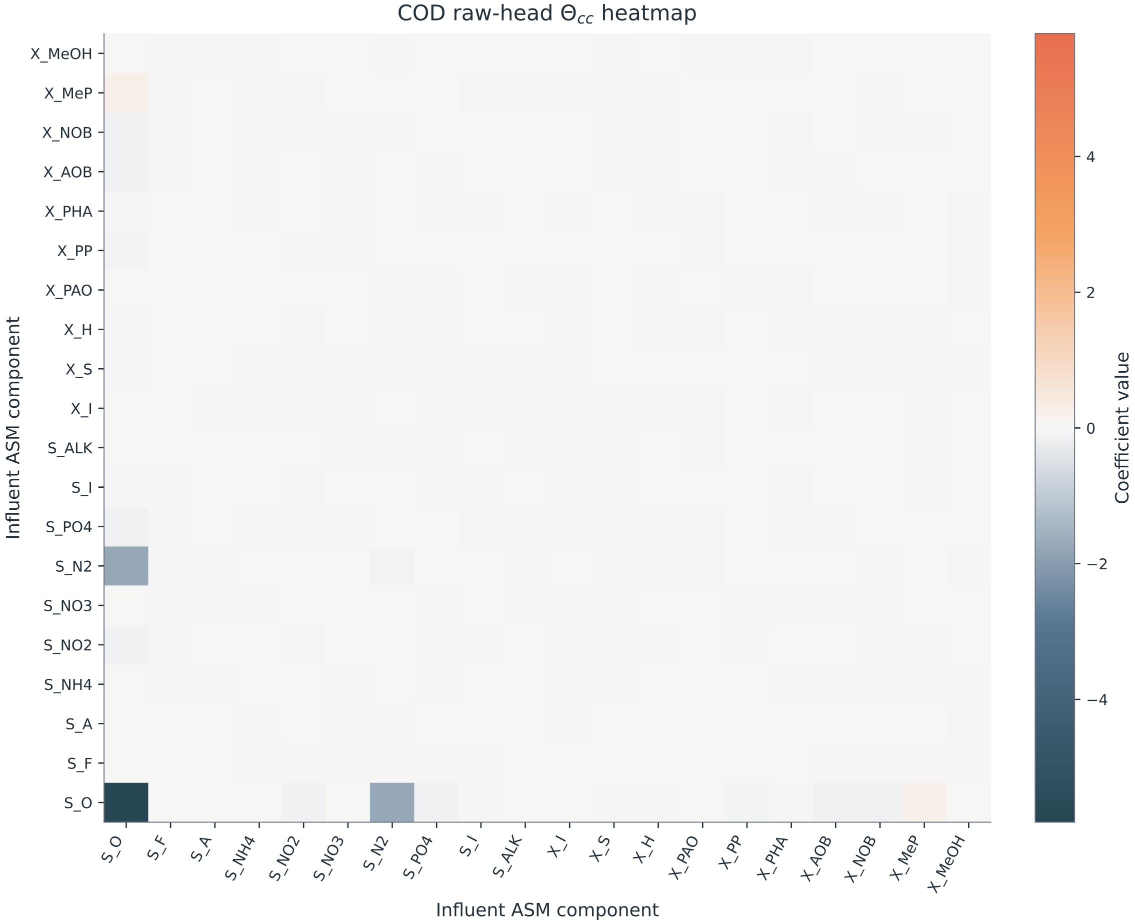

Figure 4: COD raw-head Θ_{cc} heatmap from $B_{raw,ext} = I_{comp}\hat{R}^{-1}\hat{B}$. Rows and columns enumerate influent ASM components. The matrix is symmetric to machine precision; off-diagonal cells are mirrored and each displayed cell is half the coefficient of the corresponding unordered cross term. The full raw-head atlases are reported in Supplementary Figures S1–S4.

reported-composite labels. The final affine/LP map is active-set dependent and has no single global coefficient matrix; Figure 4 must not be read as a derivative of the final deployed output.

The COD raw-head surface is exactly symmetric at stored precision. Its largest-magnitude entry is the S_O–S_O self-curvature term (−5.8097), followed by the mirrored S_O–S_{N2} pair (−1.7411). Other prominent terms link S_O with X_{MeP} (2.3190×10^{-1}), X_{AOB} (-1.3264×10^{-1}), S_{PO4} (-1.3055×10^{-1}), X_{NOB} (-1.0182×10^{-1}), and S_{NO2} (-9.4127×10^{-2}). Under the ordered Kronecker convention, a displayed off-diagonal value $q_{ij} = q_{ji}$ contributes $(q_{ij} + q_{ji})x_i x_j = 2q_{ij}x_i x_j$ to the polynomial, so cell values and combined unordered cross-term coefficients are not interchangeable. The concentration of the largest raw-head curvatures around dissolved oxygen, nitrogen-linked states, and phosphorus/metal components is consistent with the nonlinear switching and coupled nutrient-removal behavior represented in activated-sludge models (Henze et al., 2006; Stenstrom & Song, 1991; de Araujo et al., 2013; Wentzel et al., 1992). These associations describe the fitted pre-deployment map and should not be interpreted as causal effects or as global sensitivities of the final deployed state.

Table 12 counts independently interpretable terms rather than duplicated display cells. Thus Θ_{uu} has $2(2+1)/2 = 3$ unique terms and Θ_{cc} has $20(20+1)/2 = 210$, although their displayed heatmaps contain 4 and 400 cells, respectively. The reporting threshold is post-training only and does not affect model selection, fitting, or deployment. Nineteen of 20 W_{in} entries, all three unique Θ_{uu} terms, 29 of 40 Θ_{uc} terms, and 44 of 210 unique Θ_{cc} terms exceed the COD raw-head threshold; 371 of 380 off-diagonal Γ entries exceed the separately scaled coupling threshold. Overall, 469

Table 12
Retained coefficients in the COD raw-head structure. The reported-output blocks use 10^{-4} of the largest absolute COD raw-head coefficient (4.7332×10^{-3}); the shared Γ block uses 10^{-4} of its own largest off-diagonal magnitude (3.6803×10^{-5}). Symmetric quadratic totals count only unique upper-triangular terms.

Coefficient block	Total coeffs.	Retained coeffs.	% Retained
b	1	1	100.0
W_u	2	2	100.0
W_{in}	20	19	95.0
Θ_{uu}	3	3	100.0
Θ_{uc}	40	29	72.5
Θ_{cc}	210	44	21.0
Γ	380	371	97.6
Overall summary	656	469	71.5

of 656 unique candidates (71.5%) are retained. The result indicates selective influent curvature within a comparatively dense coupling structure, not a uniformly sparse model.

The COD Θ_{cc} panel serves as a representative raw-head surface, while Supplementary Figures S1–S4 preserve the full target-wise raw-head atlases. Interpretation must consider the external mapping in Eq. (21), the component coupling Γ, and the later constrained deployment stages together; inspection of a single raw-head block does not replace analysis of the full pipeline (Brunton et al., 2016; Shen et al., 2023).

This coefficient-level view differs from PLS loadings and local ANN sensitivity analysis because the ICSOR raw head exposes named global blocks for operating effects, loading effects, within-subspace curvature, and cross-component coupling. The values are shaped by invariant-aware soft training and the coupled non-negative fitted-state variable, but raw-head feasibility is not guaranteed. Exact physical admissibility is imposed afterward by constrained deployment. The coefficient blocks therefore support transparent structural inspection without implying that the raw head is itself feasible.

The coefficients in Figure 4 and Table 12 correspond to one ICSOR fit on the full training partition. Explicit symmetry resolves the ordered-pair ambiguity within Θ_{uu} and Θ_{cc}, but it does not remove other model-level identifiability limits. The joint objective over $(B, \Gamma, \widehat{C})$ is non-convex, and B and Γ can trade off in $Rc = B\phi$; different training partitions or stationary points can therefore produce similar predictions with different coefficients. Interpretation should focus on structural patterns rather than treating a single numerical coefficient as uniquely causal. Coefficient stability across resamples remains a useful future diagnostic (Brunton et al., 2016).

5. Conclusion

This study introduced Invariant-Constrained Second-Order Regression (ICSOR), an interpretable surrogate for steady-state activated sludge systems. ICSOR separates prediction from physical enforcement. Its invariant-aware regression head provides a transparent component-level prediction, and a hierarchical deployment procedure ensures that the returned state satisfies exact stoichiometric conservation and componentwise non-negativity. The procedure first checks the raw prediction, then applies a closed-form affine projection, and finally uses a linear program when non-negativity is still violated. The affine stage was sufficient for 21.8% of test cases, while the linear program handled the remaining 78.2%. Consequently, every deployed ICSOR prediction satisfied both physical requirements. By comparison, every unconstrained baseline violated conservation on all test samples, and several also predicted negative component concentrations.

These physical guarantees came with a moderate accuracy tradeoff. On the fixed 80–20 split, ICSOR produced an aggregate test RMSE of 5.98, compared with 4.38 for the MLP and 5.30 for LightGBM. These differences correspond to increases of 36.5% and 12.8%, respectively. However, ICSOR performed strongly when data were limited: it achieved the lowest mean held-out RMSE at the smallest retained dataset size and the lowest normalized area under the RMSE learning curve across the full dataset-size study. Its train–validation RMSE gap was also small at 0.22. Training was slower than for the fastest tree-based methods, but remained faster than SVR and the MLP at the largest dataset size.

Within the scope of this study, ICSOR provides a practical balance among interpretability, data efficiency, predictive accuracy, and guaranteed physical admissibility. The evidence is limited to steady-state predictions from one standalone CSTR within the sampled influent and operating ranges. The method does not claim full kinetic or thermodynamic feasibility, and its conservation guarantee depends on the chosen stoichiometric model and system boundary; unrepresented external sources or sinks therefore remain outside that guarantee. Future work should test whether the same balance is retained for more complex reactor configurations, wider operating envelopes, and surrogate-assisted optimization and control (Sadeghassadi et al., 2018; Wang et al., 2025).

CRediT authorship contribution statement

The sole author, Egberto F. Selerio Jr., contributed to conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft, writing—review & editing, visualization, supervision, project administration, and funding acquisition.

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Data availability

The complete codebase that reproduces the results presented in this study is available in the GitHub repository `https://github.com/egbertoselerio1997/icsor.git`. The repository may be cloned using the command `git clone https://github.com/egbertoselerio1997/icsor.git`.

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